

AMENDMENT NO. 3 AUGUST 2025

TO

IS 17021 : 2018 E 20 FUEL — ADMIXTURE OF ANHYDROUS ETHANOL AND GASOLINE — AS FUEL FOR SPARK IGNITED ENGINE POWERED VEHICLES — SPECIFICATION

(Page 1, clause 1) — Substitute the following for the existing:

‘1 SCOPE

1.1 This standard prescribes requirements, and methods of sampling and test methods for E20 suitable for use as a fuel in the automobile spark-ignition internal combustion engines of vehicles with BS VI emission norms.

1.2 This standard does not purport to address all of the safety problems associated with its use. It is the responsibility of the user of this specification to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.’

(Page 2, clause 5) — Delete.

[Page 3, Table 1 (see also Amendment No. 1 and Amendment No. 2)] — Substitute the following for the existing:

‘Table 1 Requirement for 20 Percent Ethanol Blended Motor Gasoline, E20 Fuel

(Clauses 4.2, 4.3, 4.3.2, 4.3.4, 4.4.2 and C-4.2)

SI No.	Characteristics	Requirements MG BS VI Grade	Method of Test, Ref to
(1)	(2)	(3)	(4)
i)	Appearance	Clear and bright, free from un-dissolved water, foreign matter and other visual impurities	Visual
ii)	Colour	Red (MG 95)	Visual inspection
iii)	Density at 15 °C, kg/m ³	720 to 775	IS 1448 (Part 16) ¹⁾ /ISO 12185/ ASTM D4052/ASTM D1298/ IP 160
iv)	Distillation:		IS 1448 (Part 18) ¹⁾ /ISO 3405/ ASTM D86
	a) Percent evaporated at 70 °C (E70 °C), percent v/v	10 to 60	
	b) Percent evaporated at 100 °C (E100 °C), percent v/v	45 to 78	
	c) Percent evaporated at 150 °C (E150 °C), percent v/v, <i>Min</i>	85	
	d) Final boiling point, °C, <i>Max</i>	210	
	e) Residue, percent by volume, <i>Max</i>	2.0	
v)	Research octane number (RON), <i>Min</i>	95 (MG 95)	IS 1448 (Part 27) ¹⁾ / ASTM D2699
vi)	Motor octane number (MON), <i>Min</i>	83 (MG 95)	IS 1448 (Part 26) ¹⁾ / ASTM D2700
vii)	Gum content (solvent washed), g/m ³ , <i>Max</i>	40	IS 1448 (Part 29) ¹⁾ / ASTM D381

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SI No.	Characteristics	Requirements MG BS VI Grade	Method of Test, Ref to
(1)	(2)	(3)	(4)
viii)	Total sulphur, mg/kg, <i>Max</i>	10	ASTM D2622 ¹⁾ /ASTM D5453/ ASTM D3120/ASTM D7039/ ASTM D7220
ix)	Lead content (as Pb), g/l, <i>Max</i>	0.005	IP 224/ASTM D5059 ¹⁾ / ASTM D3237
x)	Reid vapour pressure (RVP) at 37.8 °C, kPa, <i>Max</i>	70	ASTM D5191 (dry method)/ EN 13016/Annex A ¹⁾ (dry method)/ASTM D4953 [see footnote ²⁾]
xi)	Vapour lock index (VLI), <i>Max</i>	1 100	By calculation (VLI = 10 × RVP + 7 × E70)
xii)	Benzene content, percent by volume, <i>Max</i>	1	ASTM D5580/ASTM D6277/ ASTM D6730/ASTM D3606/ ASTM D6839/ISO 22854 ¹⁾
xiii)	Copper strip corrosion (3 h at 50 °C), <i>Max</i>	Class 1	IS 1448 (Part 15) ¹⁾ / ASTM D130
xiv)	Water tolerance of gasoline- alcohol blends, temperature for phase separation, °C, <i>Max</i> :		Annex B
	a) Other months	10	
	b) Winter ³⁾	0	
xv)	Engine intake system cleanliness	Report - MFA used	See footnote ⁴⁾
xvi)	Olefin content, percent by volume, <i>Max</i>	18	IS 1448 (Part 23) ¹⁾ / ASTM D1319/ASTM D6730/ ASTM D6839
xvii)	Oxidation stability, minutes, <i>Min</i>	360	IS 1448 (Part 28) ¹⁾ / ASTM D525/IP 40
xviii)	Aromatic content ⁵⁾ , percent by volume, <i>Max</i>	35	IS 1448 (Part 23) ¹⁾ / ASTM D1319/ASTM D6730/ ASTM D5580/ASTM D6839
xix)	Oxygen content, percent by mass, <i>Max</i>	7.7	EN 1601/EN 13132 ¹⁾ /IS 1448 (Part 181) ⁶⁾ /ASTM D4815 ⁶⁾
xx)	Ethanol content ⁷⁾ , percent v/v	20	Annex C ¹⁾ /EN 1601/ISO 6729/ ASTM D4815 ⁸⁾ /ASTM D6839 ⁸⁾ /ASTM D6730 [see footnote ⁹⁾]
xxi)	Oxygenates, percent by volume, <i>Max</i> :		ASTM D4815 ⁶⁾
	a) Ethers containing 5 or more “C” atoms per molecules such as MTBE, ETBE or TAME	15	
	b) Any other oxygenates	Not permitted	
xxii)	Motor gasoline content, percent v/v	100 – (Ethanol content and stabilizing agents/ additives)	By calculation
xxiii)	Water content ¹⁰⁾ , mg/kg, <i>Max</i>	3 000	ASTM D6304/ISO 12937/ ISO 6296
NOTES 1 No external addition of silicon, chlorine based materials and metallic additives are allowed. 2 All the test methods referred to in this standard include a precision statement. The Interpretation of results shall be based on test method.			

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- ¹⁾ In case of dispute, this method shall be the referee method.
- ²⁾ The dry vapour test method given in Annex A shall be followed.
- ³⁾ In winter (November to February) it is expected that temperature may be lower than 0 °C in the northern hilly region and hence phase separation shall not take place till - 10 °C.
- ⁴⁾ Use of multifunctional additives (MFA) is a requirement for assuring adequate fuel system and intake system cleanliness performance in engines. Refiners/marketers of motor gasoline have to ensure the MFA has proper credentials from internationally accepted test laboratories/authorities, of having passed a minimum of one of the tests in each of the two categories of deposit control performance indicated below:

Sl No.	Deposit Type	Test No.	Test Equipment	Test Method	Pass Limit [Average IVD (Weight)]
(1)	(2)	(3)	(4)	(5)	(6)
i)	Intake valve deposit (IVD)	1	BMW IVD test	ASTM D 5500-16	100 mg/valve, <i>Max</i>
		2	MB M102E IVD	CEC F-05-A-93	50 mg/valve, <i>Max</i>
		3	FORD 2.3 Litre IVD	ASTM D 6201	90 mg/valve, <i>Max</i>
		4	MB M111-IVD test	CEC F-20-A-98	50 mg/valve, <i>Max</i>
					Limit (percent injector flow loss)
				Pass limit (percent injector flow loss)	
ii)	Port fuel injector deposit (PFI)	1	Chrysler PFI test	ASTM D 5598-01 (2012)	5 percent, <i>Max</i>
		2	PFI deposit rig	ASTM D 6421	10 percent, <i>Max</i>
NOTE — Other performance tests may be added as and when they reach qualified/standard test status.					

MFAs which are certified against National Generic Certification Option as per US EPA-97 final Rule (40 CFR Part 80 Certification Standards for Deposit Control Additives) can also be used by fuel refiners/marketers at treatment levels not less than the lower additives concentration (LAC) limits, as these MFAs meet the criteria for acceptance mentioned above.

- ⁵⁾ Aromatic content relaxation and time frame, if any, for fuel processed from Assam crude, may be guided by the notifications issued by Government of India, from time to time.
- ⁶⁾ For determination of oxygen content using IS 1448 (Part 181) or ASTM D4815, pre-dilution of sample is required. Oxygen content can be calculated using the mathematical formula given in ASTM D4815 or EN 1601 after measuring all oxygenates using the test methods in col (4) of SI No. (xx) and (xxi).
- ⁷⁾ Blending of ethanol shall be carried out at supply point of Oil Marketing Company's (OMC's) depot/terminal as per the directives of Ministry of Petroleum and Natural Gas, Government of India from time to time. OMC shall ensure homogeneous blending of ethanol in motor gasoline in the range 19 percent to 20 percent for E20 fuel.
- ⁸⁾ For ASTM D4815 and ASTM D6839, dilution of sample can be done as given below:
- Dilute the E20 MG sample with oxygenate free MG in 1 : 1 ratio. Run samples as per ASTM D4815 or ASTM D6839 or ISO 22854;
 - Prepare and run one E10 MS secondary/standard reference material (SRM) or certified reference material (CRM) in same condition and note the results; and
 - Calculate ethanol percentage in E20 from following formula for results of E20 diluted samples and results of E10 CRM/SRM samples:

$$\text{Ethanol content in E20 fuel} = \left[\beta \times \frac{10}{\alpha} \right] \times 2$$

where

- β = percentage v/v ethanol in 1: 1 diluted E20 sample;
 10 = ideal SRM value, however, in case of change, actual SRM/CRM value shall be used;
 α = percentage v/v ethanol in E10 sample (SRM/CRM); and
 2 = factor obtained from dilution.

- ⁹⁾ For routine analysis of ethanol content ASTM D 5845 (FTIR) or water extraction method shall be employed (*see* Annex C).

- ¹⁰⁾ Water content shall be tested at the manufacturing location.'

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(Page 5, clause **6.2.2.1**) — Substitute the following for the existing:

‘6.2.2.1 Pump marking and labelling

Information to be marked on dispensing pumps used for delivering E20 to retail consumer. It shall be clearly visible, easily legible and displayed at any point where ethanol blended (20 percent) motor gasoline (E20) is made available to consumers. To help customers to identify E20, 20 percent EBMG BS VI shall be labelled at retail outlets as ‘E20 PETROL’.

(PCD 03)

Publication, BIS, New Delhi